

Concept Document

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Target: Concept

Privilege Escalation: When attackers gains more access or permissions than they are supposed to have on a system or Network.

Two types of Privilege Escalation:

1. Vertical Privilege Escalation
2. Horizontal Privilege Escalation

Vertical Privilege Escalation: A user gains access to higher privileges than they are supposed to have.

Example: Suppose a hacker breaks into a low level employee account than finds the flawing of system that leads them becoming an admin, now they can control everything.

Horizontal Privilege Escalation: Where the user stage at the same privilege level but accesses data or resources of another user.

Example:

Doctor A can see only his own patients.

By changing a value like doctor_id=101 to 102, he sees Doctor B's patients.

Both are normal doctors (same level)

Real World Example:

1. 🚬 Dirty Pipe (Linux Kernel) - 2022

A vulnerability in Linux kernel versions 5.8 to 5.16 allowed any normal user to become root by overwriting read-only files.

Real attack scenario:

A hacker with a simple shell account on a shared server could:

- Override /etc/passwd to create a new root user
- Modify system binaries
- Read any file (including /etc/shadow with passwords)

Companies affected:

Cloud servers, Android devices, many Linux web host

Fix: Kernel update (patch released within days)

2. 🖨 Microsoft Windows Print Spooler (PrintNightmare) - 2021

A flaw in Windows Print Spooler service allowed any authenticated user to get SYSTEM privileges.

Real attack:

- Normal domain user sends a print job
- Uses malicious printer driver installation
- Printer service (running as SYSTEM) executes arbitrary code

Result: Normal employee → Full control of Windows server

Impact: All Windows versions (Server 2016, 2019, Windows 10, 11) including fully patched systems at that time